

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer, specializing in backend development and AI-driven experiences. Possesses a Master of Science in Computer Science with a strong track record in designing and implementing scalable microservices and distributed data pipelines, leveraging expertise in Java, Python, Spark, SQL, AWS, and Azure. Demonstrates proficiency in CI/CD automation, system design, and building large-scale data platforms with a focus on privacy-preserving AI applications.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance RESTful APIs using Java 8/11 and Spring Boot, serving over 50,000 daily requests with 99.99% uptime and adhering to robust system design principles.
- Developed and automated CI/CD pipelines with Jenkins and Azure DevOps, accelerating deployment cycles across Azure Cloud and Oracle WebLogic environments for enterprise applications.
- Spearheaded a modernization initiative for legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling to reduce issue resolution time by 20 percent.
- Led a critical EDX platform upgrade, streamlining application architecture and dependency management to achieve 50% faster delivery cycles within Agile teams.
- Collaborated within Agile and Scrum frameworks to deliver FHIR-compliant healthcare solutions, utilizing Git for version control and JUnit for comprehensive testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Designed and optimized Java and EJB-based enterprise applications and RESTful web services for major automotive clients, demonstrating strong backend development skills.
- Improved system performance by 90% through meticulous SQL query optimization, database tuning, and modernization of legacy Java codebases.
- Implemented CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates to ensure high software quality.
- Mentored junior developers through comprehensive code reviews and debugging sessions, fostering clean coding practices and effective problem-solving.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99% uptime through proactive monitoring and performance tuning, showcasing a keen eye for operational stability.
- Enhanced SQL data pipelines and database operations, improving query performance by 10% and strengthening data integrity for critical business intelligence.
- Implemented comprehensive unit testing using JUnit4, achieving 99% code coverage and ensuring robust software development practices.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient project management.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, demonstrating expertise in ETL pipelines and large-scale data platforms.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, enabling effective data analysis and track record of user engagement.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments, facilitating efficient infrastructure automation.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40% through advanced system design techniques.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for scalable solutions.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, showcasing proficiency in backend development for AI experiences.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, highlighting innovative software design.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, demonstrating a commitment to responsible AI development.

- Leveraged Hugging Face and PyTorch for model development, contributing to advancements in privacy-preserving LLM-based applications.

SKILLS

- Programming Languages: Proficient in Java and Python for robust backend development and AI solutions. Expertise in SQL for complex data manipulation and relational database design.
- Frameworks and Libraries: Extensive experience with Spring Boot, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable applications and AI models. Strong command of JUnit for test-driven development.
- Web Services and APIs: Skilled in designing and implementing RESTful APIs and SOAP APIs, ensuring seamless integration and high availability.
- Cloud Platforms: Certified experience with AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps, ServiceBus) for cloud-native development and infrastructure automation.
- Databases and Data Tools: Proficient with PostgreSQL, Oracle Database, IBM DB2, and RDS. Experienced in data warehousing, ETL pipelines, and data analysis with Tableau and QuickSight.
- Development Tools and Methodologies: Adept in Docker, Kubernetes, Jenkins, Git, Maven, JIRA, Terraform, and Agile/ Scrum methodologies, with a proven track record in CI/CD and system design.